

IE2024 – Probability & Statistics
Individual Assignment (20%)

Educational Data Analysis Using Statistical Techniques and Predictive Modelling

1. Assignment Overview

The purpose of this assignment is to provide students with practical experience in applying statistical techniques to solve a real-world educational problem using an authentic dataset. Each student is required to identify an educational problem, obtain an appropriate dataset from Kaggle, perform a comprehensive statistical analysis, develop a predictive model, and communicate the findings through an academic research poster. Students will also submit a short presentation video explaining their poster and defend their work during an individual viva examination.

The assignment is designed to evaluate both statistical knowledge and the ability to interpret results in a meaningful educational context.

2. Learning Outcomes Assessed

Upon successful completion of this assignment, students should be able to:

- Apply descriptive statistical techniques to summarize data.
- Perform appropriate statistical inference techniques.
- Develop and evaluate a predictive model.
- Interpret statistical findings and communicate conclusions effectively.
- Present and defend analytical decisions professionally.

3. Dataset Requirements

Each student must select:

- An educational topic of interest.
- A publicly available dataset from Kaggle.

Examples include (but are not limited to):

- Student Performance
- Student Dropout Prediction
- Online Learning Analytics
- Student Mental Health
- Student Employability
- Academic Success
- School Attendance
- Higher Education Performance
- MOOC Learning Behaviour
- Educational Engagement

The selected dataset should:

- Contain sufficient observations
- Include both numerical and/or categorical variables
- Be suitable for statistical analysis and predictive modelling

Task 1 – Dataset Introduction

Briefly describe:

- Educational problem
- Dataset source
- Number of observations
- Number of variables
- Variable descriptions
- Target variable (if applicable)

Task 2 – Descriptive Statistical Analysis

Perform a comprehensive descriptive analysis including appropriate techniques such as:

- Frequency tables
- Measures of central tendency
- Measures of dispersion
- Percentiles or quartiles
- Correlation analysis
- Appropriate graphical visualizations

Interpret every important result.

Task 3 – Statistical Inference

Select suitable inferential statistical techniques based on your research problem.

Possible analyses include:

- Confidence intervals
- One-sample hypothesis tests
- Two-sample hypothesis tests
- Paired hypothesis tests
- ANOVA
- Chi-square tests
- Correlation significance tests

For every statistical test, clearly state:

- Research question
- Null hypothesis
- Alternative hypothesis
- Significance level

- Test statistic
- p-value
- Decision
- Interpretation

Students are expected to justify why each statistical technique was selected.

Task 4 – Predictive Modelling

Develop one suitable predictive model.

Possible models include:

- Linear Regression
- Multiple Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest
- Naïve Bayes

Your analysis should include:

- Feature selection
- Data preprocessing
- Model training
- Model evaluation
- Interpretation of results

Appropriate evaluation metrics should be reported.

Task 5 – Conclusions

Summarize:

- Major findings
- Educational implications
- Practical recommendations
- Limitations
- Possible future improvements

5. Deliverables

Students must submit the following:

1. Research Poster (PDF)

The poster should include:

- Title
- Student details
- Introduction

- Dataset
- Methodology
- Descriptive Analysis
- Statistical Inference
- Predictive Model
- Results
- Conclusions
- References

Poster Size:

A1 Portrait or Landscape

2. Source Code

Submit:

- R script (.R)

All submitted code must execute without errors.

3. Dataset

Include the original dataset or provide the Kaggle download link.

4. Poster Presentation Video

Duration:

5 minutes

The video should explain:

- Educational problem
- Dataset
- Statistical analysis
- Predictive model
- Key findings
- Conclusions

The presenter must be clearly visible throughout the presentation.

6. Assessment Scheme (20%)

Component	Weight
Poster Evaluation	50%
Individual Viva	50%

Poster Evaluation Rubric (50%)

Criterion	Marks
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Problem definition and dataset selection	5
Descriptive statistical analysis	10
Statistical inference	10
Predictive modelling	10
Interpretation of results	5
Poster organization and visual quality	5
References and academic writing	5
Total	50

Individual Viva Rubric (50%)

Students should be able to explain and justify every aspect of their work.